ASSIGNMENT 1

Structural Dynamics and Aeroelasticity

Politecnico di Milano

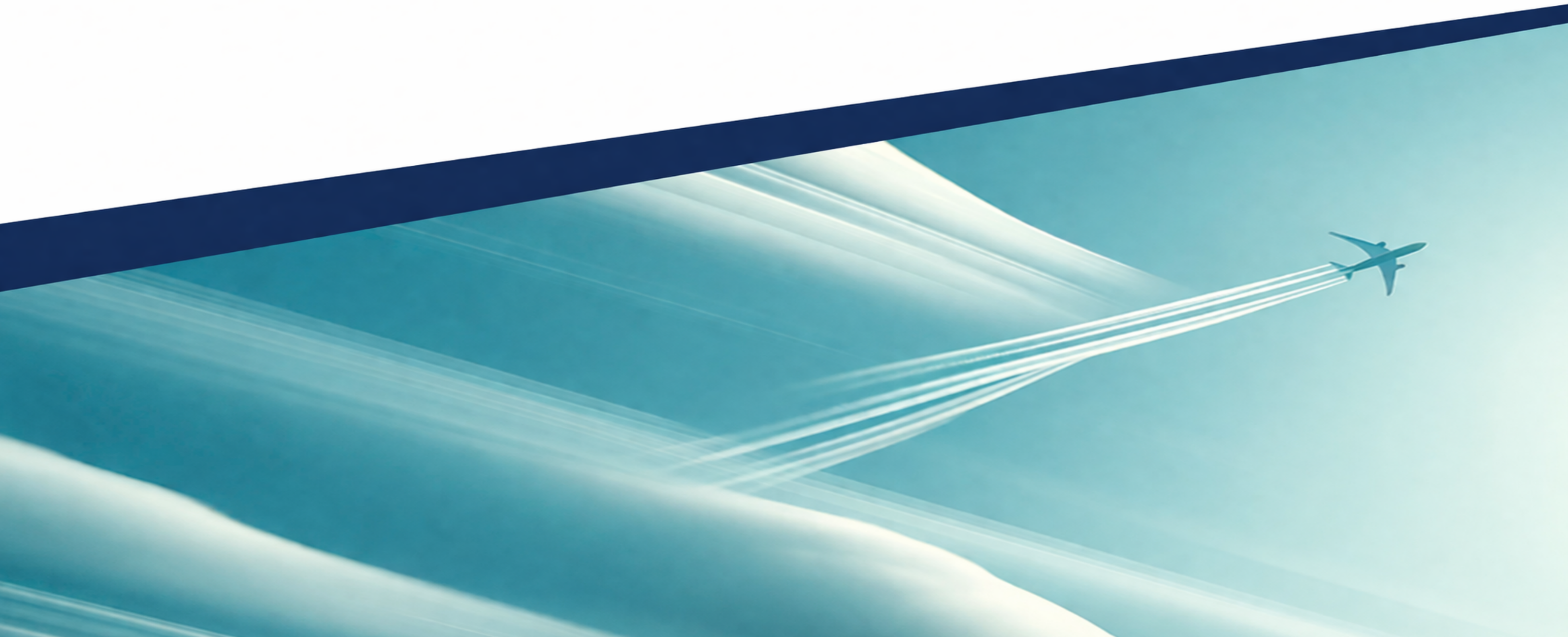
Laurea Magistrale – Aeronautical Engineering

Author : Camile Sam Adam BRIARD

Person Code : 11154383

Professors : G. Quaranta

Date : April 10, 2026



Problem Setup

The wing has two DOFs — bending $w(y)$ and torsion $\theta(y)$ — with a strut at $y = y_B$, aerodynamic center offset e from the elastic axis, and a tip aileron of span b_a and deflection β. The Ritz–Galerkin method is applied to perform a static aeroelastic analysis.

Divergence Dynamic Pressure q_D - Derivation

Shape Functions

Root clamp imposes $w(0) = w'(0) = \theta(0) = 0$. The Ritz ansatz $w = N_w(y)\,\phi_w$, $\theta = N_\theta(y)\,\phi_\theta$ requires N_w quadratic and N_θ linear. Applying BCs and normalizing to unity at the tip :

$$\boxed{N_w(y) = \left(\frac{y}{b}\right)^2, \qquad N_\theta(y) = \frac{y}{b}} \tag{1}$$

Principle of Virtual Work

In order to compute q_D, we need to find the equations of motion. To do so, we use the PVW : We need to compute the internal and external virtual work.

Internal Virtual Work

Substituting $N''_w = 2/b^2$ and $N'_\theta = 1/b$:

$$\delta W_i = \int_0^b \delta w_{,yy}\, EJ\, w_{,yy}\, dy + \int_0^b \delta\theta_{,y}\, GJ\, \theta_{,y}\, dy = \delta\phi_w \frac{4EJ}{b^3}\phi_w + \delta\phi_\theta \frac{GJ}{b}\phi_\theta \tag{2}$$

$$\boxed{\delta W_i = \delta\boldsymbol{\phi}^T \begin{bmatrix} K_w & 0 \\ 0 & K_\theta \end{bmatrix} \boldsymbol{\phi}, \qquad K_w = \frac{4EJ}{b^3}, \quad K_\theta = \frac{GJ}{b}} \tag{3}$$

External Virtual Work

$$\delta W_e = \delta W_{\text{AERO}}^{\text{wing}} + \delta W_{\text{AERO}}^{\text{aileron}} + \delta W_{\text{STRUT}} + \delta W_{\text{INERTIA}} + \delta W_{\text{GRAVITY}}$$

Since we are interested in the divergence dynamic pressure q_D, we can neglect inertia and gravity (static case, no external forces). Thus :

$$\boxed{\delta W_e = \delta W_{\text{A}}^{\text{w}} + \delta W_{\text{A}}^{\text{a}} + \delta W_{\text{STR}}}$$

(1) Wing aerodynamic virtual work δW_A^w The kinematically admissible AC displacement is $h_{AC} = w + e\,\theta$, and strip lift reads $L'_w = q\,c\,C_{L/\alpha}\,\theta$ (perturbation, $\alpha_{\text{eff}} = \theta$, $C_{L_0} = 0$). Thus :

$$\delta W_A^w = \int_0^b \delta h_{AC}\, L'_w\, dy = \int_0^b \delta\phi_w \left(\frac{y}{b}\right)^2 q\,c\,C_{L/\alpha}\,\frac{y}{b}\,\phi_\theta\, dy + \int_0^b \delta\phi_\theta\,\frac{y}{b}\,e\,q\,c\,C_{L/\alpha}\,\frac{y}{b}\,\phi_\theta\, dy \tag{4}$$

$$\boxed{\delta W_A^w = \delta \boldsymbol{\phi}^T \, q \, \mathbf{K}_A \, \boldsymbol{\phi}}, \qquad \mathbf{K}_A = \begin{bmatrix} 0 & \dfrac{cb}{4} C_{L/\alpha} \\ 0 & \dfrac{ecb}{3} C_{L/\alpha} \end{bmatrix} \tag{5}$$

$\mathbf{K}_A$ is the aerodynamic stiffness matrix.

(2) Aileron aerodynamic virtual work δW_A^a Not needed for q_D (homogeneous problem) but derived for later use. The aileron spans $[b - b_a, \, b]$ with deflection β ; perturbation gives $C_{m_{AC}} = C_{m/\beta}\beta$. The virtual work splits into a lift contribution and a pitching moment contribution :

$$\delta W_A^a = \int_{b-b_a}^{b} \delta(w + e\theta) \, q \, c \, C_{L/\beta} \, \beta \, dy + \int_{b-b_a}^{b} \delta\theta \, q \, c^2 \, C_{m/\beta} \, \beta \, dy \tag{6}$$

$$\boxed{\delta W_A^a = \delta \boldsymbol{\phi}^T \, q\beta \, \mathbf{F}_a}, \qquad \mathbf{F}_a = \frac{c}{b} \left\{ \begin{matrix} C_{L/\beta} \cdot \dfrac{b^3 - (b - b_a)^3}{3b} \\ \left(e \, C_{L/\beta} + c \, C_{m/\beta}\right) \dfrac{b^2 - (b - b_a)^2}{2} \end{matrix} \right\} \tag{7}$$

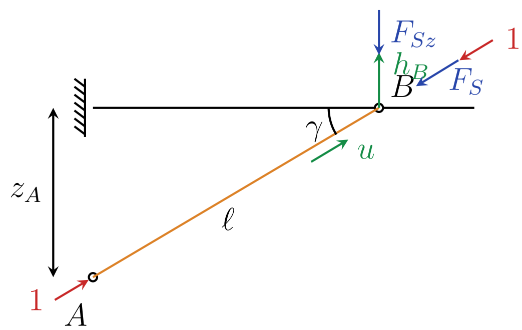


FIGURE 1 – Equivalent model of the strut, replaced by an equivalent axial spring of stiffness K_{STR}.

(3) Strut virtual work δW_{STR} The strut axial force and its z-component follow from geometry ($\ell = z_A / \sin\gamma$, $u = h_B \sin\gamma$) :

$$\boxed{F_{Sz} = -\underbrace{\frac{EA}{z_A} \sin^3\gamma}_{K_{\mathrm{STR}}} \, h_B} \tag{8}$$

The displacement at attachment point B, using $\eta_B = z_A / (b \tan\gamma)$:

$$h_B = w(y_B) - x_B \, \theta(y_B) = \eta_B \left[\eta_B \, \phi_w - x_B \, \phi_\theta\right]$$

Substituting into $\delta W_{\mathrm{STR}} = \delta h_B \, (-K_{\mathrm{STR}}) \, h_B$:

$$\boxed{\delta W_{\mathrm{STR}} = -\delta \boldsymbol{\phi}^T \, K_{\mathrm{STR}} \, \eta_B^2 \begin{bmatrix} \eta_B^2 & -\eta_B x_B \\ -\eta_B x_B & x_B^2 \end{bmatrix} \boldsymbol{\phi}} \tag{9}$$

Application of PVW : $\delta W_i = \delta W_e$

$$\delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T \begin{bmatrix} K_w & 0 \\ 0 & K_\theta \end{bmatrix} \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = \delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T \left(q\mathbf{K}_A \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} + q\,\beta\,\mathbf{F_a} - K_{\mathrm{STR}}\,\eta_B^2 \begin{bmatrix} \eta_B^2 & -\eta_B x_B \\ -\eta_B x_B & x_B^2 \end{bmatrix} \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} \right) \tag{10}$$

Assembling all contributions and invoking the arbitrariness of $\delta\boldsymbol{\phi}$ (unforced case $\beta = 0$ for q_D computation) :

$$\boxed{\left(\begin{bmatrix} K_w + K_{\mathrm{STR}}\,\eta_B^4 & -K_{\mathrm{STR}}\,\eta_B^3\,x_B \\ -K_{\mathrm{STR}}\,\eta_B^3\,x_B & K_\theta + K_{\mathrm{STR}}\,\eta_B^2\,x_B^2 \end{bmatrix} - q\,\mathbf{K}_A \right) \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}} \tag{11}$$

This is $(\mathbf{K}_S - q_D\,\mathbf{K}_A)\,\boldsymbol{\phi} = \mathbf{0}$, with :

$$\mathbf{K}_S = \begin{bmatrix} K_{S_1} & K_{S_{12}} \\ K_{S_{12}} & K_{S_2} \end{bmatrix}, \qquad \mathbf{K}_A = \begin{bmatrix} 0 & \mathbf{K_{A_1}} = \dfrac{cb}{4}\,C_{L/\alpha} \\ 0 & \mathbf{K_{A_2}} = \dfrac{ecb}{3}\,C_{L/\alpha} \end{bmatrix} \tag{12}$$

$$K_{S_1} = \frac{4EJ}{b^3} + \frac{EA\,z_A^3}{b^4}\,\frac{\cos^4\gamma}{\sin\gamma}, \qquad K_{S_{12}} = -\frac{EA\,z_A^2\,\cos^3\gamma\,x_B}{b^3}, \qquad K_{S_2} = \frac{GJ}{b} + \frac{EA\,\sin\gamma\,\cos^2\gamma\,z_A\,x_B^2}{b^2} \tag{13}$$

For divergence, the condition is : $\det(\mathbf{K}_S - q_D\,\mathbf{K}_A) = 0$. Expanding the determinant :

$$K_{S_1}K_{S_2} - q_D\,K_{S_1}\,K_{A_2} - K_{S_{12}}^2 + q_D\,K_{S_{12}}\,K_{A_1} = 0 \quad \Rightarrow \quad \boxed{q_D = \frac{K_{S_1}\,K_{S_2} - (K_{S_{12}})^2}{K_{S_1}\,K_{A_2} - K_{S_{12}}\,K_{A_{12}}}} \tag{14}$$